2.13.7 Main Transmission Oil Temperature

Minimum during start .... –30 °C

Normal operation range .... +30 to +105 °C

EFFECTIVITY *If extreme low temperature kit 105-80006 installed and operational*

Normal operation range .... 0 to +105 °C

EFFECTIVITY *ALL*

Maximum .... +105 °C

Transient (10-min limited) .... +120 °C

2.14 FUEL LIMITATIONS

2.14.1 Fuel Pressure

Normal operating range .... 0.6 to 1.7 kp/cm^2

Maximum .... 1.7 kp/cm^2

2.14.2 Fuel Specifications

Fuel conforming to the following specification is authorized for use:

Type Fuels (Primary)

Wide-cut Type	- MIL-T-5624, grade JP-4	*)
	- ASTM D-1655, Jet B	
	- CAN 2-3.22	
Kerosene Type	- MIL-T-5624, grade JP-5	*)
	- MIL-T-83133, grade JP-8	*)
	- ASTM D-1655, Jet A or Jet A1	
	- CAN 2-3.23	
	- GOST #10227, TS-1 (Regular or Premium)	
	- GOST #16564, RT	

*) Contains an anti-icing additive which conforms to MIL-I-27686 and does not require additional anti-icing additives.

Type Fuels (Emergency)

CAUTION MIL-G-5572 FUEL CONTAINING TRICRESYLPHOSPHATE (TCP) ADDITIVE SHALL NOT BE USED.

MIL-G-5572E and subsequent, all grades without TCP

NOTE Maximum of 6 hours operation per turbine overhaul period.

2.14.3 Fuel Specifications - Cold Weather

WARNING AT AMBIENT TEMPERATURES BELOW 4 °C, SOME TYPE OF FUEL ANTI-ICING ADDITIVE IS REQUIRED.

To assure consistent starts at ambient temperatures below 4 °C, the following fuels are recommended:

Type Fuels (Primary)

Wide-cut Type	- MIL-T-5624, grade JP-4	*)
	- ASTM D-1655, Jet B	

*) Contains an anti-icing additive which conforms to MIL-I-27686 and does not require additional anti-icing additives.

NOTE
- Jet A, Jet A1, JP-5 or JP-8 may start the engines at temperatures below 4 °C; however, when cold soaked, marginal starts may result due to viscosity changes. Once started, the engines will operate satisfactorily on JP-5, JP-8, Jet A and Jet A1 at fuel temperature and OAT down to –32 °C.
- GOST specification fuels must have a minimum viscosity of 6 centistokes when starting cold engines under low temperature conditions.

Type Fuels (Alternate)

CAUTION
- JP-4 OR JET B FUEL SHALL NOT BE MIXED WITH AVGAS.
- THIS HELICOPTER MAY NOT BE OPERATED WITH AN AVGAS/JET FUEL MIXTURE WHEN THE OAT IS 4 °C OR HIGHER.

AVGAS/Jet A, Jet A1, JP-5 or JP-8 (MIL-T-83133) mixture

The AVGAS/Jet fuel mixture is defined as:

– One part by volume AVGAS (MIL-G-5572) of either grade 80/87 or grade 100/130 (100L) having 0.53 ml/l maximum lead content. Do not use grade 100/130 having 4.6 ml/l lead content.

– Two parts by volume ASTM D-1655, Jet A or Jet A1, or MIL-T-5624, grade JP-5.

NOTE Use of 100/130 (100L) grade AVGAS/Jet fuel mixture shall be restricted to 300 hours in one engine overhaul period due to the high lead content of the fuel.

2.14.4 Fuel Additives

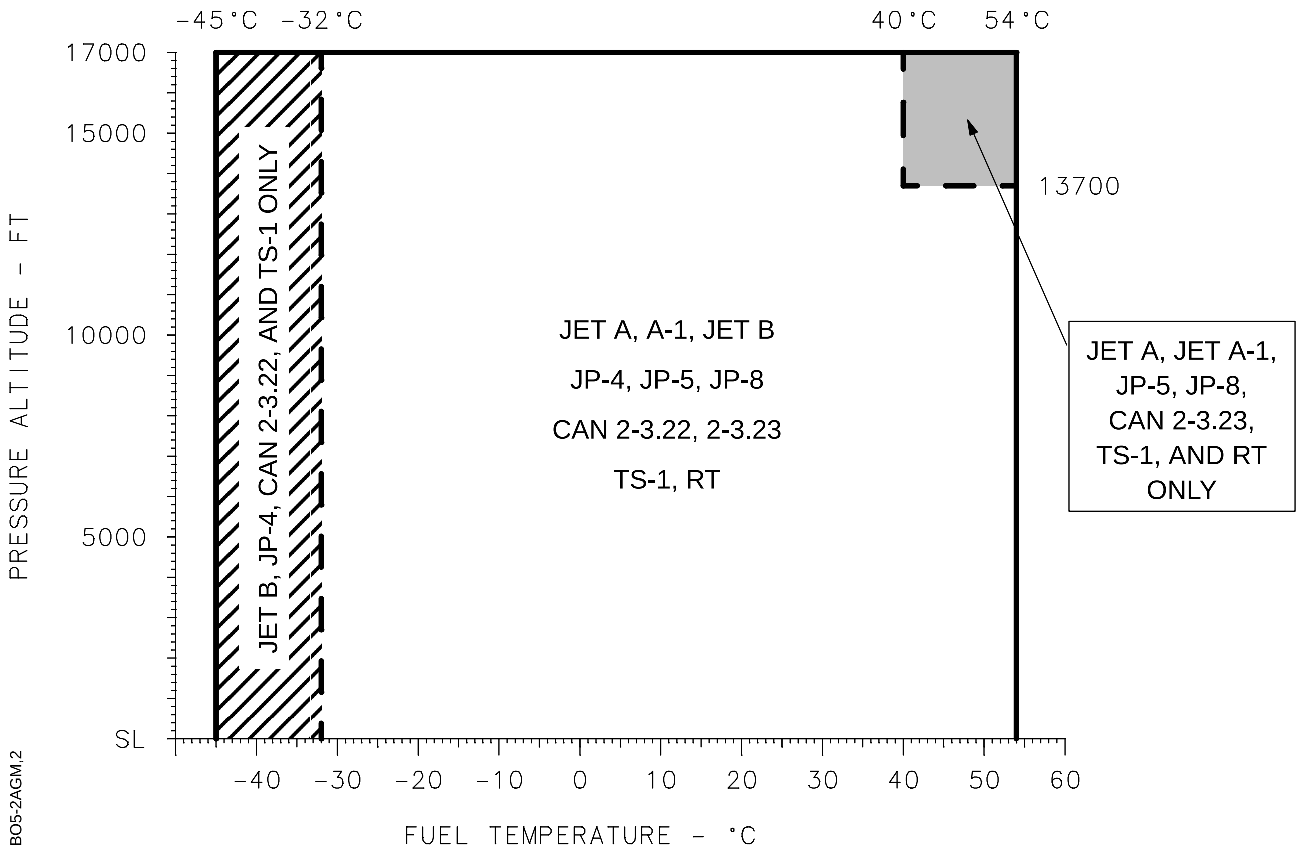
2.14.4.1 Anti-icing

Icing protection is required when operating with primary fuels at temperatures below +4 °C (40 °F).

– MIL-I-27686; Concentration by volume: Max 0.15 %, min 0.035 %.
– Fluid I (GOST 8313); Concentration by volume: Max 0.3 %, min 0.10 %. *)
– Fluid IM (TU 6-10-1458); Concentration by volume: Max 0.3 %, min 0.10 %. *)

*) Only for use with GOST specification fuels.

2.14.5 **Fuel Temperature**



NOTE REFER TO HELICOPTER OAT LIMITS.

Fig. 2-4 Temperature Limits for Approved Primary Fuels

2.14.6 **Fuel Quantities**

TANK	**USABLE FUEL**		**UNUSABLE FUEL**	
	liters	kilograms	liters	kilograms
Main	477	382	7.5	6.0
Supply	93	74	2.5	2.0
Totals	570	456	10.0	8.0

Fuel mass values are based on a fuel density of 0.8 kg/l.

The usable fuel quantity is empty when zero is indicated on the relevant fuel quantity gauge.